WSABI Summer Research Lab **Summer 2026**

Lecture 15: Regularization and Ridge Regression

Instructor: Jonathan Pipping-Gamón *Authors: JP, RB*

15.1 Measuring Player Impact in Basketball

Basketball teams want to know how much each player contributes to winning. At the possession level, that usually comes down to two things: helping the offense score and helping the defense prevent points. That suggests several possible measurements:

- **Box-score production**: points, assists, rebounds, steals, blocks, turnovers, and shooting efficiency.
- **Raw plus-minus (P/M)**: how the score changes while the player is on the court.
- **On-off plus-minus**: how the team performs with the player on the court versus off the court.

Each measure captures part of the story, but none isolates the player cleanly:

- Box scores miss invisible contributions such as spacing, screen setting, ball pressure, and help defense;
- Raw plus-minus gives a player credit or blame for everyone sharing the court;
- On-off numbers still depend on substitution patterns, bench units, opponent quality, and garbage-time lineups.

The central confounding problem is *who plays with whom*. A bench player who shares most of his minutes with stars can look better than he is. A strong defender assigned to tough opponent lineups can look worse than he is. APM is one way to ask for a player-impact estimate after accounting for the other nine players on the court.

15.2 Adjusted Plus-Minus

Index possessions by $i = 1, \ldots, n$. Let y_i be the point margin for the offensive team on possession i, measured in points per possession or scaled to points per 100 possessions. Let

$$OP_1(i), \ldots, OP_5(i)$$

be the offensive players and

$$DP_1(i), \ldots, DP_5(i)$$

be the defensive players.

One useful model gives every player two latent contributions:

- β_j^{off}: offensive contribution of player j;
- β_j^{def}: defensive contribution of player j, written so larger is better defense.

Then

$$y_i = \alpha_0 + \beta^{\text{off}}_{OP_1(i)} + \cdots + \beta^{\text{off}}_{OP_5(i)} - \beta^{\text{def}}_{DP_1(i)} - \cdots - \beta^{\text{def}}_{DP_5(i)} + \epsilon_i, \qquad \mathbb{E}[\epsilon_i] = 0.$$

The minus sign means that better defenders reduce the offense's scoring margin.

Equivalently, build a design matrix $\boldsymbol{X}$ with one column for the intercept, one offensive indicator column for each player, and one defensive indicator column for each player. For row i,

$$x_i = [1, \overset{\text{Player 1 off}}{\bullet}, \ldots, \overset{\text{Player } m \text{ off}}{\bullet}, \overset{\text{Player 1 def}}{\bullet}, \ldots, \overset{\text{Player } m \text{ def}}{\bullet}],$$

where offensive bullets are 1 for offensive players on the court, defensive bullets are -1 for defensive players on the court, and all other entries are 0. The model is

$$\boldsymbol{y} = \boldsymbol{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}.$$

For some purposes, the offense-defense split is useful. For a compact comparison of P/M, APM, and RAPM, we can also collapse each player into one total-impact coefficient:

$$x^{\text{total}}_{ij} = \begin{cases} 1, & \text{player } j \text{ is on offense on possession } i, \\ -1, & \text{player } j \text{ is on defense on possession } i, \\ 0, & \text{otherwise.} \end{cases}$$

Then β_j is a one-number plus-minus effect: positive values mean the player's team tends to score more, and allow fewer points, after the model accounts for the other players on the floor. The same OLS-versus-ridge issue appears in either version of the design matrix.

Fitting this model by ordinary least squares gives APM. OLS chooses the coefficients with the smallest total squared prediction error:

$$\widehat{\boldsymbol{\beta}}^{OLS} = \arg\min_{\boldsymbol{\beta}} \sum_{i=1}^{n} \left(y_i - x_i^T \boldsymbol{\beta}\right)^2.$$

15.3 Why APM Needs Regularization

OLS has one job: explain the observed possessions as well as possible. Basketball lineup data make that dangerous because they are not clean experiments.

- **Teammates move together.** Starting groups and bench groups play many possessions as units, so their columns in $\boldsymbol{X}$ are highly correlated.
- **Some players have little data.** Low-minute players can receive extreme estimates from a small number of noisy stints.
- **Roles are not random.** Coaches choose who plays with stars, who guards elite scorers, and who closes games.
- **Possessions are noisy.** Made threes, missed free throws, transition chances, and end-of-clock shots create randomness that is not player skill.

In matrix terms, $\boldsymbol{X}^T\boldsymbol{X}$ can be close to singular. Small changes in the data can then produce large changes in the fitted coefficients. A player can look like an all-league defender or a replacement-level player because a few lineups ran hot or cold.

The model contains many related player effects, and OLS is willing to assign noisy variation to individual coefficients when that improves in-sample fit.

15.4 Ridge Regression

Ridge regression changes the fitting objective. Instead of only minimizing residual error, it also penalizes large coefficients:

Definition 15.1 (Ridge Regression)**.** *For a fixed $\lambda > 0$, the ridge estimator is*

$$\widehat{\boldsymbol{\beta}}^{Ridge} = \arg\min_{\boldsymbol{\beta}} \left\{ \sum_{i=1}^{n} \left(y_i - x_i^T \boldsymbol{\beta} \right)^2 + \lambda \sum_{j=1}^{p} \beta_j^2 \right\}.$$

The intercept is usually left unpenalized.

The first term measures fit to the observed possessions. The second term charges the model for large player coefficients.

The tuning parameter λ controls how much shrinkage we apply:

- $\lambda = 0$: ordinary APM, fit by OLS;
- small λ: mild shrinkage toward zero;
- large λ: strong shrinkage, so most player effects stay near league average.

The closed-form ridge solution is

$$\widehat{\boldsymbol{\beta}}^{Ridge} = \left(\boldsymbol{X}^T \boldsymbol{X} + \lambda \boldsymbol{I} \right)^{-1} \boldsymbol{X}^T \boldsymbol{y}.$$

The term $\lambda \boldsymbol{I}$ adds a positive ridge to the diagonal of $\boldsymbol{X}^T \boldsymbol{X}$. Intuitively, this makes the unstable directions of the design matrix harder to chase. If two teammates always share the floor, OLS can wildly split credit between them; ridge prefers a smaller, more conservative split unless the data strongly justify a large separation.

When APM is fit with ridge, it is often called **Regularized Adjusted Plus-Minus (RAPM)**.

15.5 Choosing the Amount of Regularization

A useful λ is chosen by held-out prediction, not by whether the resulting leaderboard feels right.

A standard workflow is:

1. Split the data into training, validation, and test sets.
2. Choose a grid of candidate values, such as $\lambda \in \{10^{-3}, 10^{-2.5}, \ldots, 10^{3}\}$.
3. For each λ, fit the RAPM model on the training set.
4. Evaluate each fitted model on the validation set.
5. Select the λ with the best validation performance.
6. Evaluate on the test set once the final model has been chosen.

The validation set influences model choice. If one λ predicts validation possessions better than another, that is evidence for choosing it. The test set has a different role: it is a final audit of the selected procedure. If you use a test set to choose λ, it is no longer an honest estimate of future performance.

In basketball data, splits should match the prediction problem. If the goal is to predict future possessions, a time-based split is natural: fit on earlier games, validate on later games, and reserve the final chunk of games as the test set. Cross-validation can replace a single validation set when data are limited, but the same principle remains: the final test set comes after the model class, features, and λ selection rule are locked.

15.6 Comparing Plus-Minus Estimates

The main visual signature of RAPM is shrinkage. Compared with raw P/M and unregularized APM, the ridge estimates stay closer to zero unless the lineup evidence is strong.

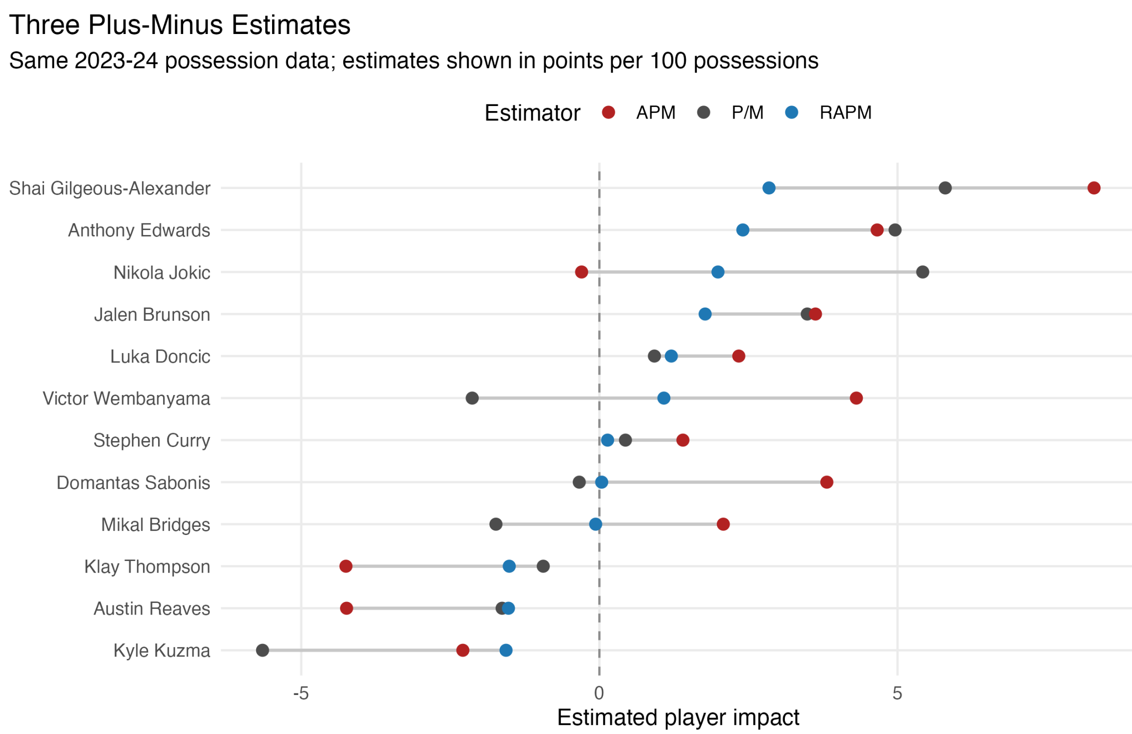

Figure 15.1: Raw P/M, APM, and RAPM estimates for selected 2023–24 NBA players. Estimates are in points per 100 possessions and centered at league average.

Shrinkage is useful only if it improves future predictions, not just if the estimates look more reasonable. Holdout evaluation tests that directly on games the model did not see. Here, raw P/M and APM actually predict *worse* than the mean-only baseline, while RAPM predicts better.

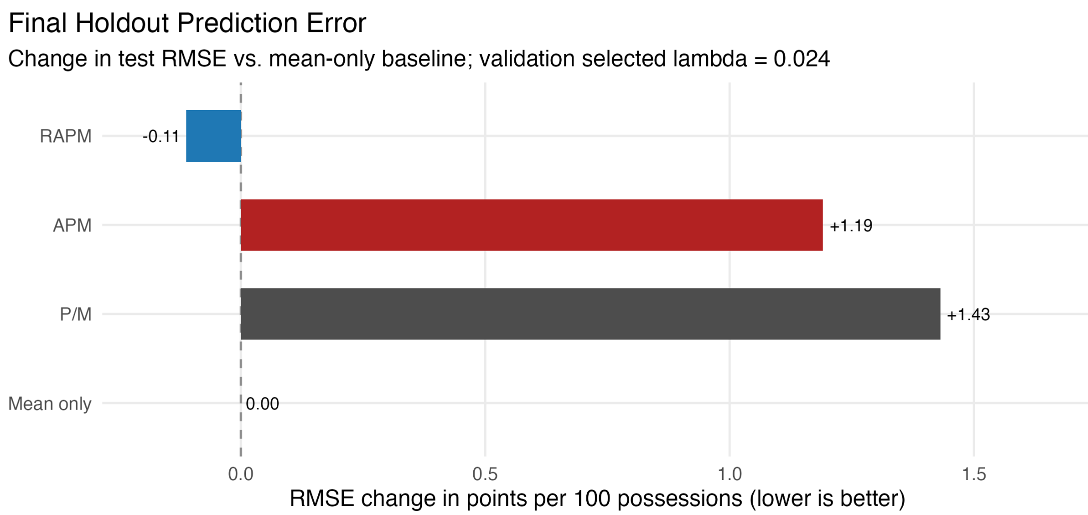

Figure 15.2: Holdout RMSE relative to a mean-only baseline. Lower is better.

Ridge regression deliberately introduces bias by pulling player effects toward zero. Its benefit, however, is lower variance. For player evaluation, this is usually a good trade: we keep some signal about which players appear better or worse, but we do not overreact to noise in the data. This keeps players with limited minutes from getting extreme estimates and keeps the model from assigning arbitrary credit splits among players in tightly linked lineups. With the right λ, out-of-sample prediction usually improves.

15.7 Ridge as a Normal Prior

The ridge penalty has a Bayesian interpretation. Suppose the likelihood is Gaussian:

$$\boldsymbol{y} \mid \boldsymbol{\beta} \sim \mathcal{N}\left(\boldsymbol{X}\boldsymbol{\beta}, \sigma^2 \boldsymbol{I}\right).$$

Now put an independent zero-centered normal prior on each player coefficient:

$$\beta_j \sim \mathcal{N}(0, \tau^2).$$

The maximum a posteriori estimate, or MAP estimate, is the value of $\boldsymbol{\beta}$ with the largest posterior probability. Equivalently, it minimizes the negative log posterior.

The negative log posterior is, up to constants,

$$-\log \mathbb{P}(\boldsymbol{\beta} \mid \boldsymbol{X}, \boldsymbol{y}) = \frac{1}{2\sigma^2} \sum_{i=1}^{n} \left(y_i - x_i^T \boldsymbol{\beta}\right)^2 + \frac{1}{2\tau^2} \sum_{j=1}^{p} \beta_j^2.$$

Multiplying by $2\sigma^2$ gives

$$\sum_{i=1}^{n} \left(y_i - x_i^T \boldsymbol{\beta}\right)^2 + \frac{\sigma^2}{\tau^2} \sum_{j=1}^{p} \beta_j^2.$$

Multiplying an objective by a positive constant does not change which $\boldsymbol{\beta}$ minimizes it. Therefore the MAP estimate solves

$$\widehat{\boldsymbol{\beta}}^{MAP} = \arg\min_{\boldsymbol{\beta}} \left\{ \sum_{i=1}^{n} \left(y_i - x_i^T \boldsymbol{\beta}\right)^2 + \frac{\sigma^2}{\tau^2} \sum_{j=1}^{p} \beta_j^2 \right\}.$$

This is the ridge objective with

$$\lambda = \frac{\sigma^2}{\tau^2}.$$

So ridge is the **maximum a posteriori** estimate under a normal prior centered at zero, with the prior variance determining the penalty strength. In this view, a large positive or negative player-impact estimate needs enough lineup evidence to overcome a prior centered at league average.

A smaller prior variance τ^2 means we believe player impacts are tightly clustered around average, so shrinkage is stronger. A larger τ^2 means we are more willing to believe large player differences, so shrinkage is weaker.

15.8 Takeaways

- APM estimates player contributions by adjusting for the other players on the court.
- OLS APM is unstable when lineups are collinear and player samples are small.
- Ridge regression stabilizes APM by shrinking player effects toward zero.
- λ should be chosen with validation data or cross-validation, not by inspecting the final test set.
- Ridge is equivalent to a MAP estimate under a zero-centered normal prior on player effects.